

Rsearch

A user-friendly R interface to VSEARCH with integrated visualization and automated parameter tuning

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MOTIVATION

VSEARCH¹ is a fast and versatile open-source bioinformatics tool for metabarcoding analysis. Its speed and compatibility have made it a popular choice for high-diversity environmental DNA (**eDNA**) datasets, where it has been shown to outperform tools such as DADA2 and Deblur ². However, its command-line interface may **limit accessibility** for many researchers. To address this, we developed **Rsearch**, an **R-based interface** to VSEARCH with additional tools for **visualization** and **automated parameter tuning**.

Rsearch OVERVIEW

Rsearch supports key workflow steps including **trimming**, **filtering**, **merging**, **dereplication**, **chimera removal**, **denoising**, **clustering**, and **taxonomic classification**. In addition, it extends VSEARCH with tools for visualization, parameter tuning, and data conversion.

KEY FEATURES

- I . Streamlined workflow in R
- II . Automated parameter tuning
- III . Integrated visualization
- IV . Modular use of selected components
- V . Flexible data handling
- VI . Easy integration with other R packages

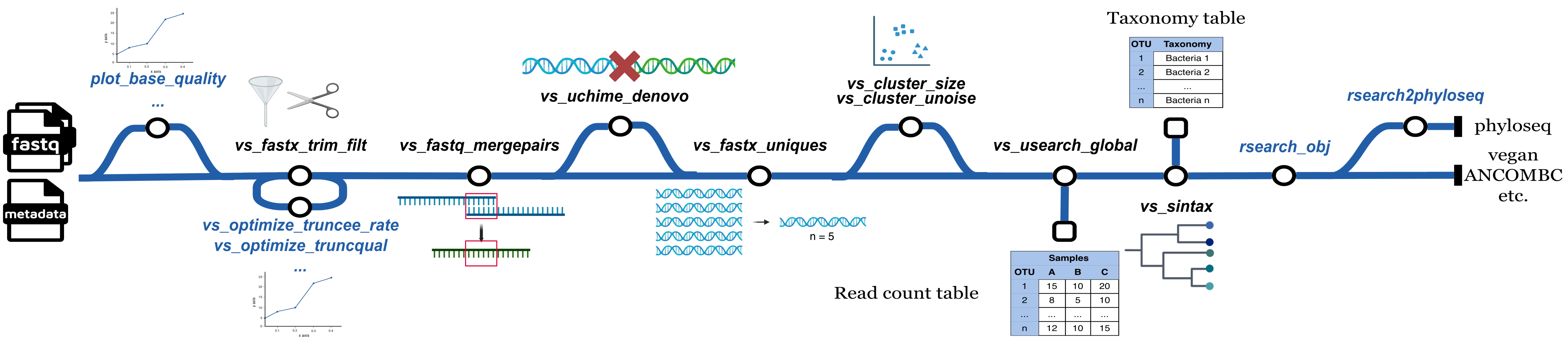


Figure 1. Example workflow for paired-end sequence analysis with *Rsearch*. The main path represents a typical metabarcoding pipeline, while detours indicate optional steps for visualization or automated parameter tuning.

Per-position quality scores: median and mean



Figure 2. Inspect paired-end read quality with *plot_base_quality*. The shaded regions indicate where the read-pairs overlap each other, and important information with respect to 3' end trimming. This is a new feature of typical quality profile plots.

4066 merged pairs at rate 0.01 (total 10000, size > 2)

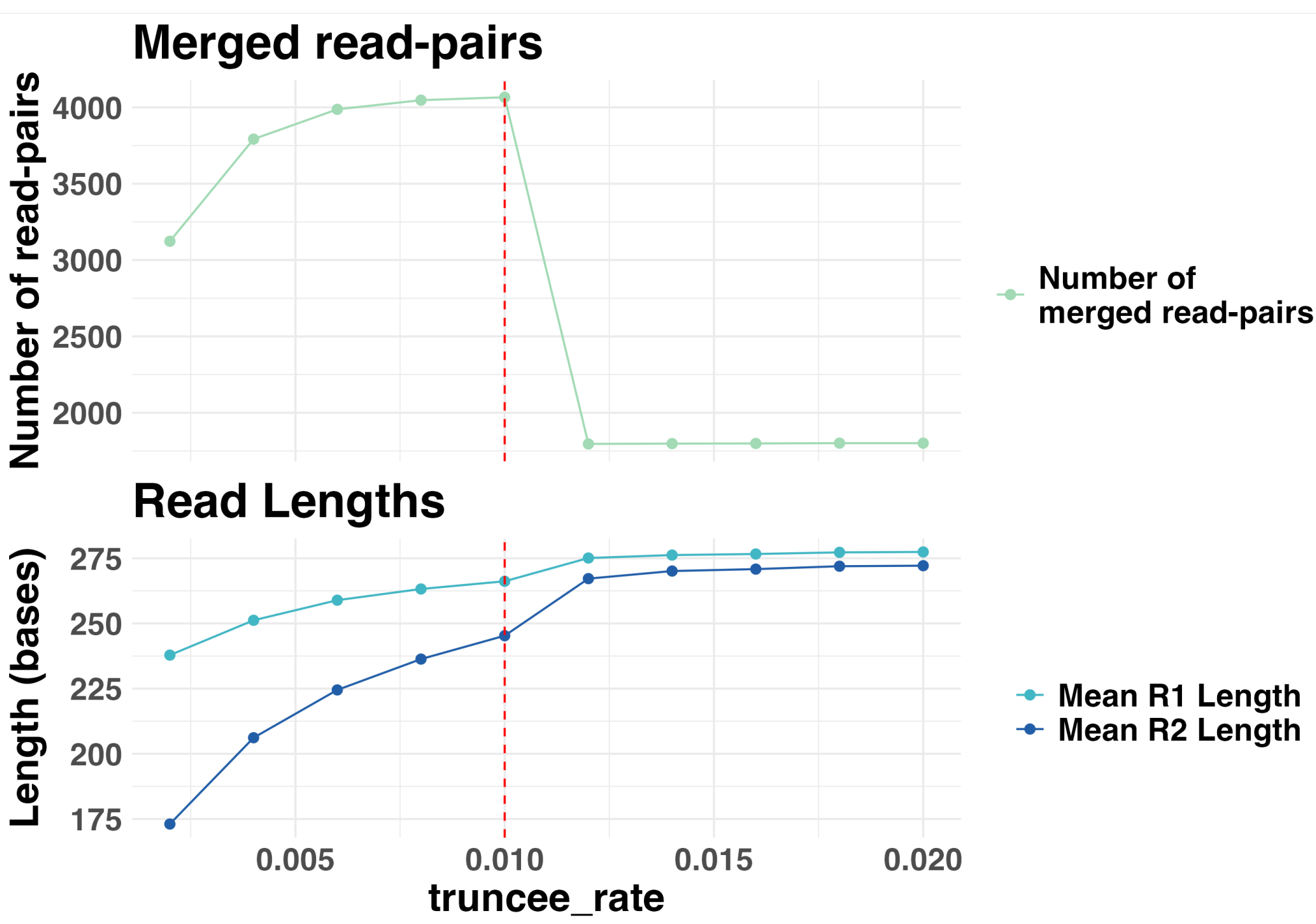


Figure 3. Automated tuning with *vs_optimize_trunceer_rate*. *truncation_rate* truncates the 3' end of reads to keep the average expected error per base below a specified threshold. The red vertical line marks the optimal value, defined as the one that maximizes merged read pairs with copy number ≥ 2 , assuming sequencing errors are more frequent in singletons.

BENCHMARKING

Benchmarking against **DADA2³** on a 16S rRNA mock community dataset showed that both pipelines produced **accurate relative abundance profiles**. However, **diversity estimates differed**: DADA2 consistently over-split the sequences, inflating low-prevalence OTUs. In contrast, **Rsearch** produced fewer, more prevalent, and more stable OTUs, indicating a more conservative and robust approach.

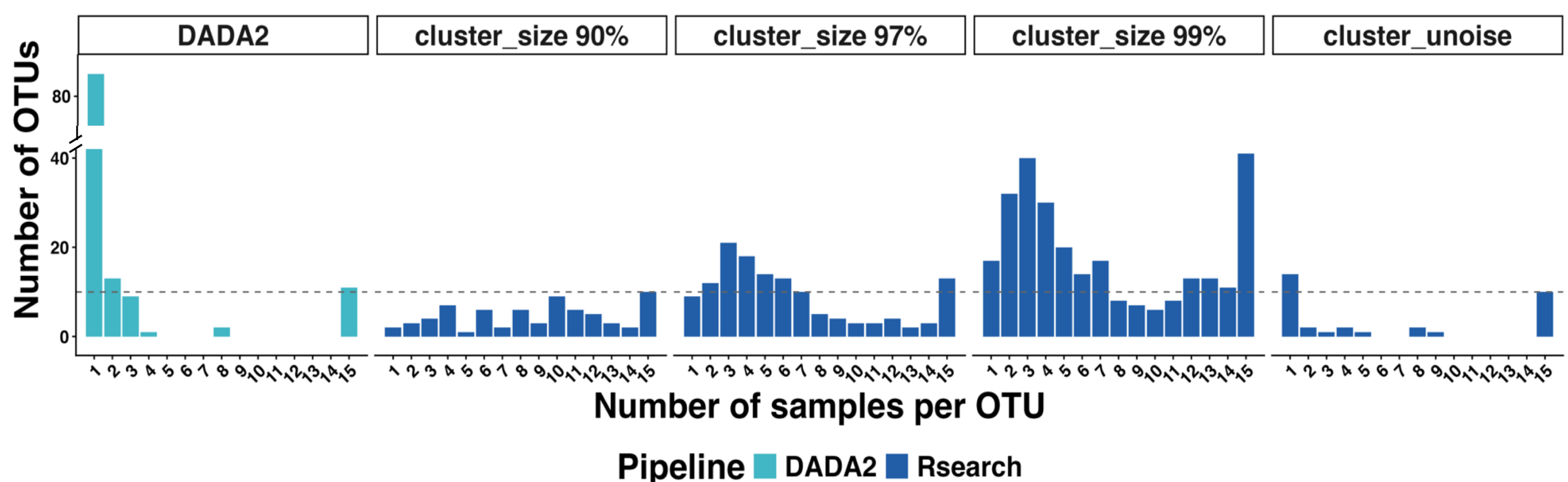
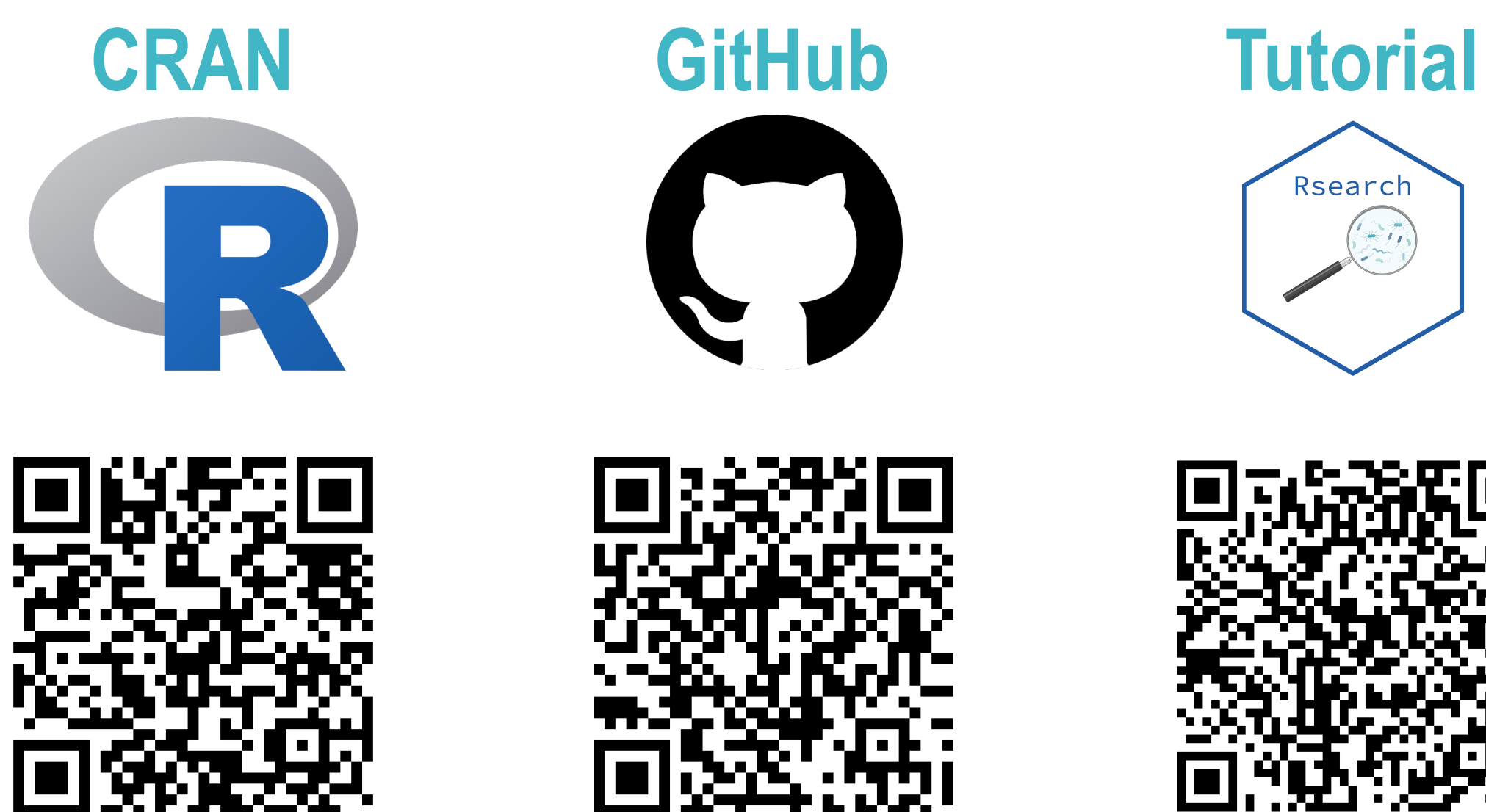


Figure 4. OTU sample prevalence. The horizontal dashed line indicates the expected 10 OTUs present across all 15 mock community samples.

CONCLUSIONS & AVAILABILITY

Rsearch combines the efficiency of VSEARCH with the accessibility and flexibility of R, providing a reproducible and user-friendly framework for metabarcoding analysis. Its automated parameter tuning, together with integrated visualization, streamlines the workflow and positions it as a potential key tool for eDNA metabarcoding analysis, particularly for high-diversity datasets.



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